Performance Test Results

Final deliverable 9 — SENG 34213 System Development Project

Hotel Website (HW), Forever Hotel. Snapshot: 2026-10-25 (Asia/Colombo).

Assessment status: Tooling verification completed; NFR-14 / 300-user staging acceptance NOT EXECUTED

Source revision: `63673ba704b4ae645a8f067e246309b6fb0f03d0`. Branch: `docs/190-release-setup-and-demo-evidence`.

The reports were prepared after this commit; they do not certify a deployed release.

Scope is the HW repository plus identified API Gateway evidence, not the complete multi-subsystem hotel system.

Guideline source: SENG34213_System_Development_Project (1).pdf, section 10.1, printed page 40. The guideline suggests nested testing/security paths; these files are directly under `documents/` as requested by the user.

1. Objective

Measure website room search/booking responsiveness, photos, calendars and quote refresh under coordinated system traffic, and demonstrate last-room protection. The backlog's DDP-#34 / GitHub #188 requires a realistic 300-user system-wide run with infra and agreed NFR-14 thresholds. No staging URLs, approved numeric thresholds, allocated hardware or real staging dataset were supplied for this report.

2. Tests actually executed

Command: `npm run test:coverage` from performance on 2026-10-25; Node v22.14.0, win32. Eight disposable HTTP fixture scenarios passed in 7.373 seconds of test-runner wall time. This is not application response time or a throughput/capacity benchmark.

Scenario	Observed outcome / interpretation
percentiles and threshold failures include slow requests, transport errors and payload sizes	PASS — local fixture assertion; no live capacity claim
configuration refuses missing targets, duplicate guests and overlapping last-room inventory	PASS — local fixture assertion; no live capacity claim
mixed HTTP load measures every operation, checks persisted race winner and cleans up without reporting credentials	PASS — local fixture assertion; no live capacity claim
fails the report for overbooking	PASS — local fixture assertion; no live capacity claim
fails the report for imageFailure	PASS — local fixture assertion; no live capacity claim
fails the report for cleanupFailure	PASS — local fixture assertion; no live capacity claim

Scenario	Observed outcome / interpretation
fails the report for timeout	PASS — local fixture assertion; no live capacity claim
fails the report for malformedReservation	PASS — local fixture assertion; no live capacity claim

Metrics module coverage: 100% line/branch/function. Runner coverage: 94.34% lines, 90.30% branches, 100% functions. These measure test-tool code execution, not the number of hotel guests the system supports.

3. Actual load results availability

Measure	Result
Staging URL / deployed digest	Not supplied / not verified
Approved NFR-14 p95 / error budget	Not supplied; example config deliberately rejects null targets
Coordinated system users / duration	300 requested by backlog; no staging load executed
Actual p50/p95/p99 / RPS / error rate	Not measured for deployed application
Photo payload/cache measurements	Implemented in runner; no actual staging cold/warm result
Gateway/Manager timeout impact	Counters/reporting implemented; not measured against deployed upstreams
Real SQL last-room contention under load	CI test exists; not executed in this report
Hardware/replicas/DB pool/dataset	Not supplied; record before run
Bottleneck / NFR-14 pass/fail	Undetermined; no passing acceptance claim

4. Runner workload and metrics

The closed-loop runner supports up to 300 virtual guests with ramp/think time and a distinct precreated test account per user. It fetches HTML, one configured original/optimized photo pair per cycle, catalogue/promotions, availability/calendar, quote and refreshed quote, then periodically creates/cancels an unpaid reservation. A separate one-room stay is reserved concurrently by two guests. Passing that check requires one winner, one ROOM_UNAVAILABLE conflict, no remaining availability and a persisted owned pending status before cleanup.

The JSON records per-operation p50/p95/p99/max, status counts, error fraction, client timeouts, received decoded body bytes, image ratio, throughput and generator metadata. Every required operation, agreed threshold, race and acknowledged cleanup must pass. It never starts live checkout or sends a payment callback. Stalled requests, 502/504, 429, failed cleanup and ambiguous reservation outcomes must remain visible.

5. Limitations

Fixture contention is not proof of real PostgreSQL locking. The current DB HTTP integration attempt is blocked by connectivity; existing pooled SQL concurrency tests are gated to disposable localhost CI. A website-only 300-user run is not a complete multi-subsystem hotel-system test. Closed-loop guests are not requests per second.

Original-plus-optimized comparison adds traffic; this is not a browser cache/LCP simulation. Received body bytes are decoded payload, not on-wire compressed transfer bytes.

6. Staging execution protocol

- Infra provides isolated staging, the approved system traffic split/targets, real room inventory and hardware/dataset/revision notes.
- Configure performance/config.local.json from the example and one test account per user; keep the last-room stay isolated with exactly one available physical room.
- Establish cold/warm photo cache runs and monitoring for application, gateway, Manager, database locks/pool/CPU/RAM and generator resources.
- Run `npm run load -- config.local.json results/staging-cold.json`, then the agreed warm run. Use only dedicated test inventory because reservations are real writes.
- Inspect p95/error thresholds and timeout distributions; verify last-room state and pendingBookingIds/uncertainReservations; reconcile any unacknowledged holds.
- Attach actual sanitized JSON, monitoring timestamps, hardware/dataset, bottleneck analysis, CI real SQL contention evidence and reviewer sign-off.

See complete performance runbook. This is the next evidence step, not work already performed.

7. Conclusion

The test tooling and its negative-case detection pass locally. System responsiveness, the NFR-14 target and 300-user capacity remain unverified until the staging protocol runs. No fabricated latency, zero-error result or performance approval is included.

Evidence and sign-off

Current requirement/screen mapping: release traceability.

Prepared from executable results and source inspection; reviewer/supervisor acceptance is not supplied.

Do not attach real environment files, passwords, JWTs, card data or raw provider responses.